Last week:

- Measuring connectivity:
  - Edge-cuts, Vertex-cuts
  - Vertex- and edge-connectivity
- Menger's theorem.

This week:

- Eulerian graphs
- Hamiltonian graphs

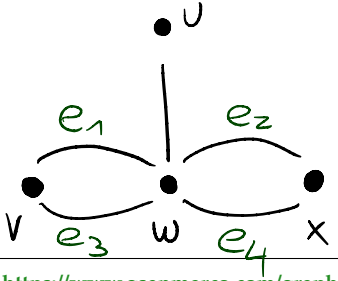
Recall: A **multigraph** might have multiple edges between any pair of vertices. Loops are not allowed.

- The degree of a vertex v in a multigraph G is the number of *edges* incident to v.
- A **walk** in a multigraph G is a collection

$$(v_0, e_1, v_1, e_2, \ldots, v_{k-1}, e_k, v_k)$$

  where $v_0, v_1, \ldots, v_k$ are vertices, $e_1, \ldots, e_k$ are edges and the endpoints of e_i are v_{i-1} and v_i.
- The length of such walk is k, the number of edges.
- A **trail** in G is a walk that traverses each edge at most once.
- A **circuit** is a trail where the first and last vertex are equal. $v_0 = v_k$
- A trail with distinct endpoints is called **open**.

Example.



deg v = 2 (only one neighbor), deg w = 5

$(v, e_1, w, e_4, x) \neq (v, e_3, w, e_4, x)$

$(v, e_1, w, e_2, x, e_4, w, e_3, v)$ circuit

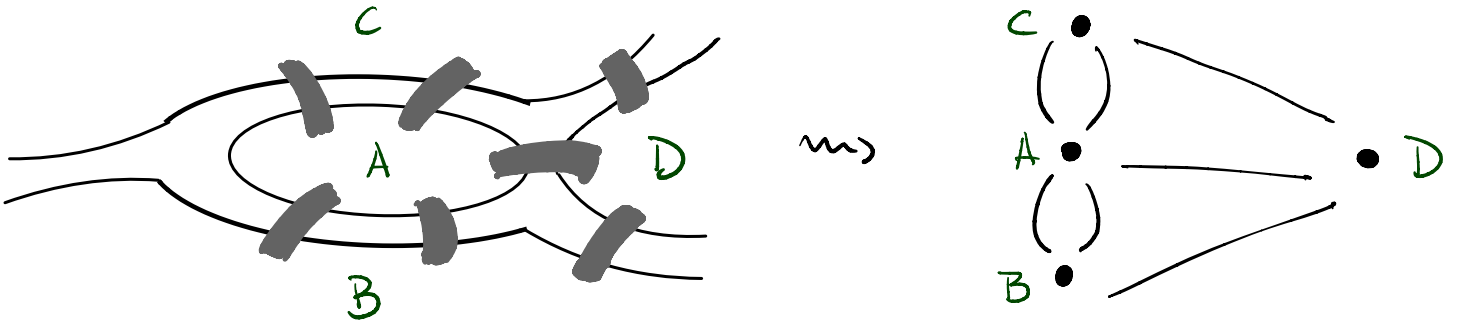
$(v, e_1, w, e_4, x, e_2, w)$ open trail

Definition. *Let* G *be a multigraph.*

- *An* ***Eulerian circuit*** *is a circuit in* G *that traverses each edge exactly once and uses all vertices.*
- *An* ***Eulerian trail*** *is an open trail in* G *that traverses each edge exactly once and uses all vertices.*
- *A graph is called* ***Eulerian*** *if it contains an Eulerian circuit.*

$(v_0, e_1, v_1, e_2, v_2, \ldots, e_k, v_k = v_0)$

Example (Seven bridges of Königsberg)**.**



Question: Is the previous multigraph Eulerian? More generally, when is a multigraph Eulerian?
Idea: If Eulerian, every time we arrive at a vertex with an edge, we leave it with the *next* edge.
Thus every vertex should have even degree.

(*) **Lemma.** *Let* G *be a multigraph, and let*

$$T = (v_0, e_1, v_1, e_2, \dots, v_{k-1}, e_k, v_k)$$

be a trail in G. *Fix* v *any vertex in* T.

(1) *If* T *is a circuit (i.e.* $v_0 = v_k$*), then* T *contains an even number of edges incident to* v.

(2) *If* T *is open and* $v \neq v_0, v_k$, *then* T *contains an even number of edges incident to* v.

(3) *If* T *is open and* $v = v_0, v_k$ *then* T *contains an odd number of edges incident to* v.

Proof: Fix v. Let $T_v = \{$edges in T incident to $v\}$

$I_v = \{e_i : 1 \le i \le k,\ v = v_i\}$ "edges going into v"

$O_v = \{e_i : 1 \le i \le k,\ v = v_{i-1}\}$ edges going out of v

Then $T_v = I_v \cup O_v$ (T doesn't repeat vertices, loops not allowed)

Hence $|T_v| = |I_v| + |O_v|$

Consider $f: E(T) \longrightarrow E(T)$, $f(e_i) = e_{i+1}$ (interpret $e_{k+1} = e_1$)

Note: f is bijective

$f^{-1}: E(T) \longrightarrow E(T)$, $f^{-1}(e_j) = e_{j-1}$ (interpret $e_0 = e_k$)

2) If $V \neq V_0, V_k$, Then $f: I_V \longrightarrow O_V$ is a bijection (There's no wraparound), Thus $|I_V| = |O_V|$

Then $|T_V| = |I_V| + |O_V| = 2|I_V|$ even!

1) Case 1.1: $V \neq V_0, V_k$ use the previous argument

Case 1.2: $V = V_0 = V_k$

$f: I_V \longrightarrow O_V$ is bijective, since $e_k \in I_V$ and $f(e_k) = e_1 \in O_V$

Then again $|T_V| = 2|I_V|$ even.

(& e_1, V_k)

3) $V_0 \neq V_k$

3.1) $V = V_k$. Here $e_k \in I_V$, but $f(e_k) = e_1 \notin O_V$

So $f: I_v \longrightarrow O_v \cup \{e_1\}$, Thus $|I_v| = |O_v| + 1$

Then $|T_v| = |I_v| + |O_v| = 2|O_v| + 1$ odd

3.2) $v = v_0$. Here $e_1 \in O_v$, but $f^{-1}(e_1) = e_k \notin I_v$

$f: I_v \cup \{e_k\} \longrightarrow O_v$, $|O_v| = |I_v| + 1$

Then $|T_v| = 2|I_v| + 1$ odd.



Consequence: in an Eulerian graph, all vertices have even degree.

Königsberg is not Eulerian

(X*) **Lemma.** *Let* G *be a multigraph where all vertices have even degree. If* v *is a non-isolated vertex, then there is a circuit in* G *using* v *and containing at least one edge.*

Proof: Let $T = (v = v_0, e_1, v_1, e_2, \ldots, e_k, v_k)$ be a maximum-length trail in G starting at v. (v not isolated $\Rightarrow k \geq 1$)

Claim: $v_k = v_0$, hence T is circuit.

Assume, for a contradiction, $v_k \neq v_0$. By lemma (*) (3) T has an odd number of edges incident to v_k

Hence there is some $e_{k+1} \in E(G) \setminus E(T)$ incident to v_k.

if v_{k+1} denotes the other endpoint, then

$T' = (v = v_0, e_1, v_1 \ldots, e_k, v_k, e_{k+1}, v_{k+1})$ contradicts T maximal

Thus $v_k = v_0$ ∎

Theorem (6.1). *Let* G *be a multigraph. Then* G *is Eulerian if and only if* G *is connected and all vertices have even degree.*

Proof: $\Rightarrow$) If G contains an Eulerian circuit, the circuit contains all vertices, thus G is connected.

The circuit uses each edge of G exactly once, thus by lemma (*) part (1), all vertices have even degree

$\Leftarrow$) Let $C = (v_0, e_1, v_1, \ldots, e_k, v_k)$ circuit of G of maximum length (we have circuits by lemma (**))

WTS: C is Eulerian circuit: i.e. $E(C) = E(G)$

Assume $E(C) \neq E(G)$. Let $G' = G - E(C)$

Claim 1: All vertices in G' have even degree

$$\deg_{G'} v = \deg_G v - \deg_C v \quad \text{even}$$

even — even — lemma (*) (1)

Claim 2: There is some $v^* \in V(C)$ which is not isolated in G'

- Case 1: $V(C) = V(G)$, clear, because $E(G) \setminus E(C)$ non-empty
- Case 2: $V(C) \neq V(G)$.

  We have $V(G) = V(C) \sqcup (V(G) \setminus V(C))$

  ($V(C)$: non-empty)

But G connected, Thus there is some $e^* \in E(G)$ going between $V(C)$ and $V(G) \setminus V(C)$.

Let v^* denote the endpoint of e^* in $V(C)$, not isolated in G' because of e^*,

By Claims 1, 2, we can apply lemma(**) to v^* in G'

To get a circuit C' in G' using V^*

Wlog $V^* = v_0$

$G - E(C)$ = G'

So $C' = (V^* = w_0, f_1, w_1, \ldots, f_\ell, w_\ell = w_0)$ in G'

Now C followed by C' is a circuit in G longer than C, contradiction.

Hence $E(C) = E(G)$ ▨

Theorem (6.2)**.** *Let* G *be a multigraph. Then* G *has an Eulerian trail if and only if* G *is connected and has exactly two odd-degree vertices.*

Exercise. *When is* K_n *Eulerian? When is* $K_{s,t}$ *Eulerian?*

Exercise. *When does* $K_{2,t}$ *contain an Eulerian trail?*